

```
align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[subtitle] linebreak() text(fill: black70, size: 6.8pt)[Five native noisy fluorescence fields. Quantitative
summary combines analytical white-noise gain with a dark-background gradient-stability proxy.] ]
v(8pt) image-grid(data, [Gradient magnitude, native fluorescence images], "magnitude_path", show-
input: true)
```

```
align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Orientation maps. Hue encodes edge angle modulo  $\pi$  and brightness follows normalized
magnitude.] ] v(8pt) image-grid(data, [Gradient orientation, native fluorescence images],
"orientation_path", show-input: false)
```

```

align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Analytical and empirical stability summary. The empirical proxy is the median absolute deviation of
gradient magnitude on the darkest percentile pixels.] ] v(8pt) grid( columns: 2, column-gutter: 10pt,
[ bar-chart(data, "white_noise_gain", [Analytical white-noise gain], [log10 WNG], log-scale: true) ],
[ bar-chart(data, "background_gradient_mad_mean", [Background gradient MAD], [MAD], log-scale:
false) ], ) v(10pt) legend(data) v(10pt) summary-table(data)

```

```

align(center)[ text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[WVF radius-trace follow-up on the fixed fluorescence image set.] linebreak() text(fill: black70, size:
6.8pt)[Garnet trace points sweep WVF across  $(r, d)$  =
(3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11). Thin horizontal lines are the fixed baseline methods.
Outlined WVF points beat the best fixed baseline on that metric.] ] v(8pt) grid( columns: 2, column-
gutter: 10pt, row-gutter: 10pt, [ wvf-trace-chart(data, "white_noise_gain", [Analytical WNG versus
WVF radius], [log10 WNG], log-scale: true) ], [ wvf-trace-chart(data,
"background_gradient_mad_mean", [Background gradient MAD versus WVF radius], [MAD], log-
scale: false) ], [ wvf-trace-chart(data, "background_gradient_median_mean", [Background gradient
median versus WVF radius], [Median], log-scale: false) ], ) v(8pt) legend(data)

```

align(center)[text(fill: black90, size: 11pt, weight: "bold")[title] linebreak() text(fill: black70, size: 8pt)
[Fine-grained WVF radius and degree sweep on the fixed fluorescence image set.] linebreak()
text(fill: black70, size: 6.8pt)[The heatmap shows native-noise background-gradient MAD across the
feasible (r , d) grid. Crossed cells were skipped by the Section 7 conditioning gate. The outlined cell
is the annotated optimum.] v(8pt) wvf-grid-heatmap(data, [Background gradient MAD across the
feasible WVF grid]) v(10pt) align(left)[text(fill: black90, size: 7.2pt, weight: "bold")[Annotated
optimum.] text(fill: black90, size: 7.2pt)[$r=9$, $d=3$] text(fill: black90, size: 7.2pt)[with MAD 0.000133.]
linebreak() text(fill: black90, size: 7.2pt, weight: "bold")[Driver assessment.] text(fill: black90, size:
7.2pt)[the native-noise fluorescence optimum is intermediate at $r=9$, which is consistent with a
compromise between fine transition scale and the modality's empirical noise floor.]]